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The Machiavellian Sandbox (MASA): Emergent Political Behavior in LLM-Driven Multi-Agent Systems

Problem Statement

Non-player characters (NPCs) in modern simulations follow rigid, pre-scripted logic trees. They cannot reason strategically, form genuine alliances, or adapt their loyalties based on emerging social dynamics. This project addresses a deeper question: *can autonomous LLM agents, constrained by game-theoretic incentive structures, produce macro-political behavior without any of it being explicitly programmed?*

Proposed system

- A political simulation where a player is dropped into a village populated by autonomous agents.
- Each agent has a hidden agenda, a persistent memory, and limited resources.
- Agents must decide each turn — whether to cooperate with the player, form coalitions with other NPCs, or betray existing alliances to monopolize resources.
- No behavior is scripted. All of it emerges from incentive structures and language-based reasoning.

The system has three layers:

- **World layer** — a shared game state tracking resources, events, and time
- **Agent layer** — each NPC runs an LLM decision loop bounded by a dynamic payoff matrix; agents negotiate with each other in natural language
- **Infrastructure layer** — a vector memory store per agent, a social graph tracker measuring trust deltas, and an emergence logger for research analysis

Agents reason in natural language but *cannot act outside their computed payoff matrix*. This grounds the LLM's outputs in game-theoretic logic rather than letting it freewheel and hallucinate
Since the layers are separate, we are not limited to just one simulation type. Combination of worlds with the agents and their behaviour is endless

Relationship with OpenClaw:

Where OpenClaw provides the plumbing for deploying agents, MASA provides the apparatus for studying them — with deterministic tick execution, game-theoretic payoff constraints, and comprehensive emergence logging unavailable in conversational systems

Outcome

- A playable demo: a village simulation where emergent alliances, rivalries, and betrayals unfold in real time with no scripted events
- A research dataset: logged social graph trajectories across 50–100 simulations under varying parameters
- A research paper: Do LLMs Discover Nash Equilibria or get stuck in defection; do LLM agents under resource scarcity and game-theoretic constraints exhibit coalition formation patterns consistent with classical political theory?